Phishing Awareness Simulation Using GoPhish

Author

Hassan Sajid

Program

Internee.pk Cybersecurity Internship

Tool Used

GoPhish - Open-Source Phishing Simulation Framework

Introduction

Phishing attacks are one of the most common forms of social engineering used by cybercriminals to compromise user accounts and gain unauthorized access to organizational systems. Attackers typically send deceptive emails that trick users into clicking malicious links or submitting sensitive credentials.

The objective of this assignment was to simulate a phishing attack in a controlled lab environment in order to understand how phishing campaigns work and how organizations can train employees to identify and avoid such attacks.

The simulation was conducted using the open-source phishing framework GoPhish. This tool allows security teams to design phishing emails, create landing pages, launch campaigns, and analyze user interaction metrics.

All activities in this lab were performed in a virtual environment for educational purposes only.

Objectives

The main objectives of this assignment were:

- Understand how phishing attacks are structured
- Deploy and configure the GoPhish phishing simulation framework
- Design a phishing email template
- Create a phishing landing page to capture credentials
- Launch a phishing campaign
- Analyze the results and user interaction metrics

This exercise helps demonstrate how security awareness programs can be implemented inside organizations.

Tools and Technologies Used

The phishing simulation environment consisted of the following tools:

Kali Linux
Used as the main attack machine to host the GoPhish server.

GoPhish
An open-source phishing framework used to create phishing campaigns and track user behavior.

VirtualBox
Used to create and manage virtual machines in the lab environment.

Windows Virtual Machine
Used as a simulated employee workstation to receive and interact with the phishing email.

SMTP Email Service
Used to send phishing emails from the GoPhish server to target users.

Lab Environment Setup

The lab environment was built using a virtualized infrastructure. The GoPhish server was deployed on Kali Linux, while a Windows virtual machine acted as the victim workstation.

Network communication between the machines was configured using VirtualBox network adapters.

The Kali Linux machine hosted the phishing server and administrative dashboard, while the Windows VM was used to receive the phishing email and simulate user interaction.

This isolated lab environment ensured that the phishing simulation could be performed safely without affecting real systems.

Installing and Running GoPhish

The GoPhish framework was downloaded and extracted on the Kali Linux machine.

After extraction, the phishing server was started using the following command:

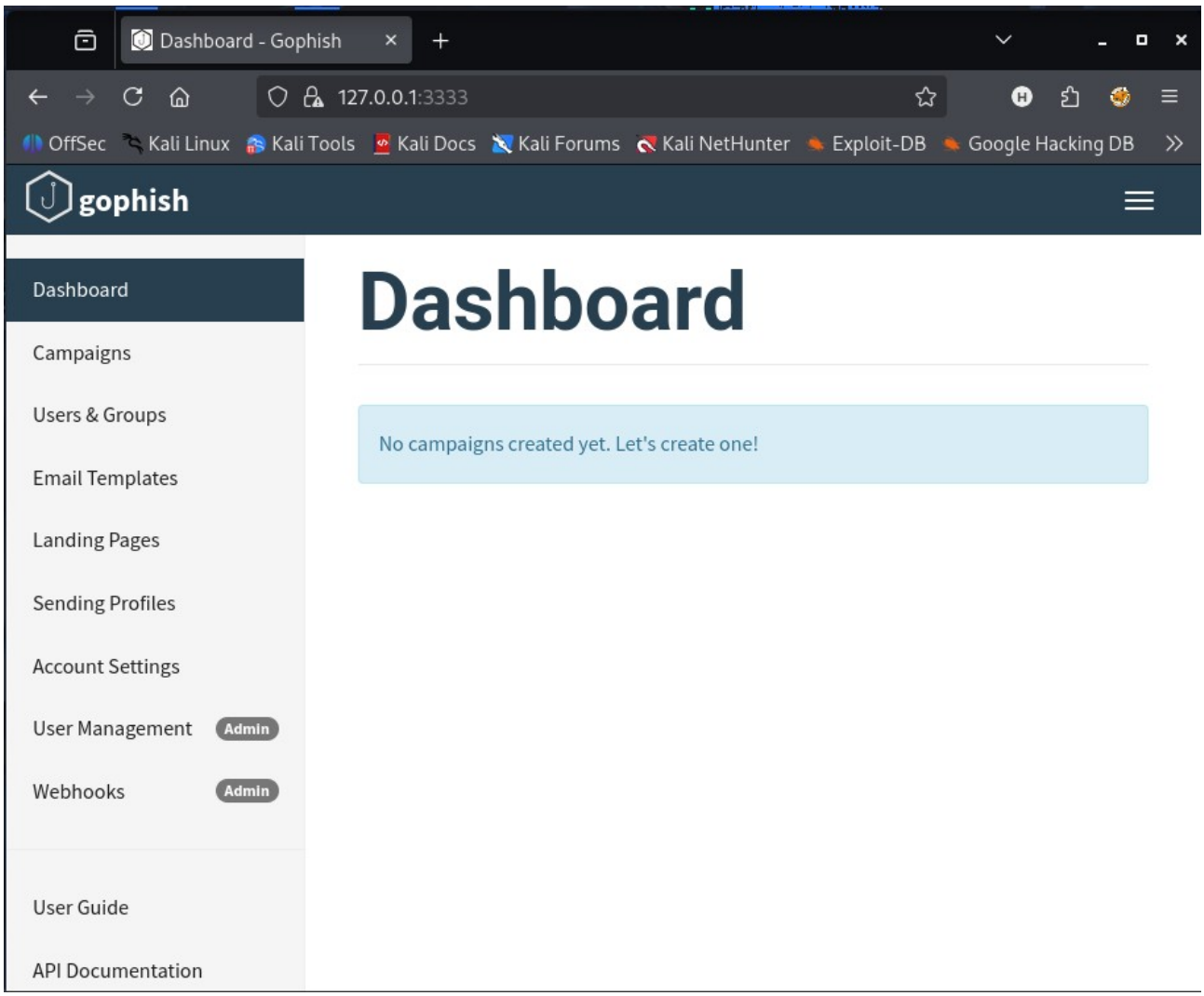
```
./gophish
```

When executed, GoPhish launched two services:

Admin Dashboard
https://127.0.0.1:3333

Phishing Server
http://0.0.0.0:80

The administrative dashboard allowed management of email templates, landing pages, campaigns, and analytics.



SMTP Configuration

To send phishing emails, SMTP settings were configured in the GoPhish dashboard.

The following parameters were configured:

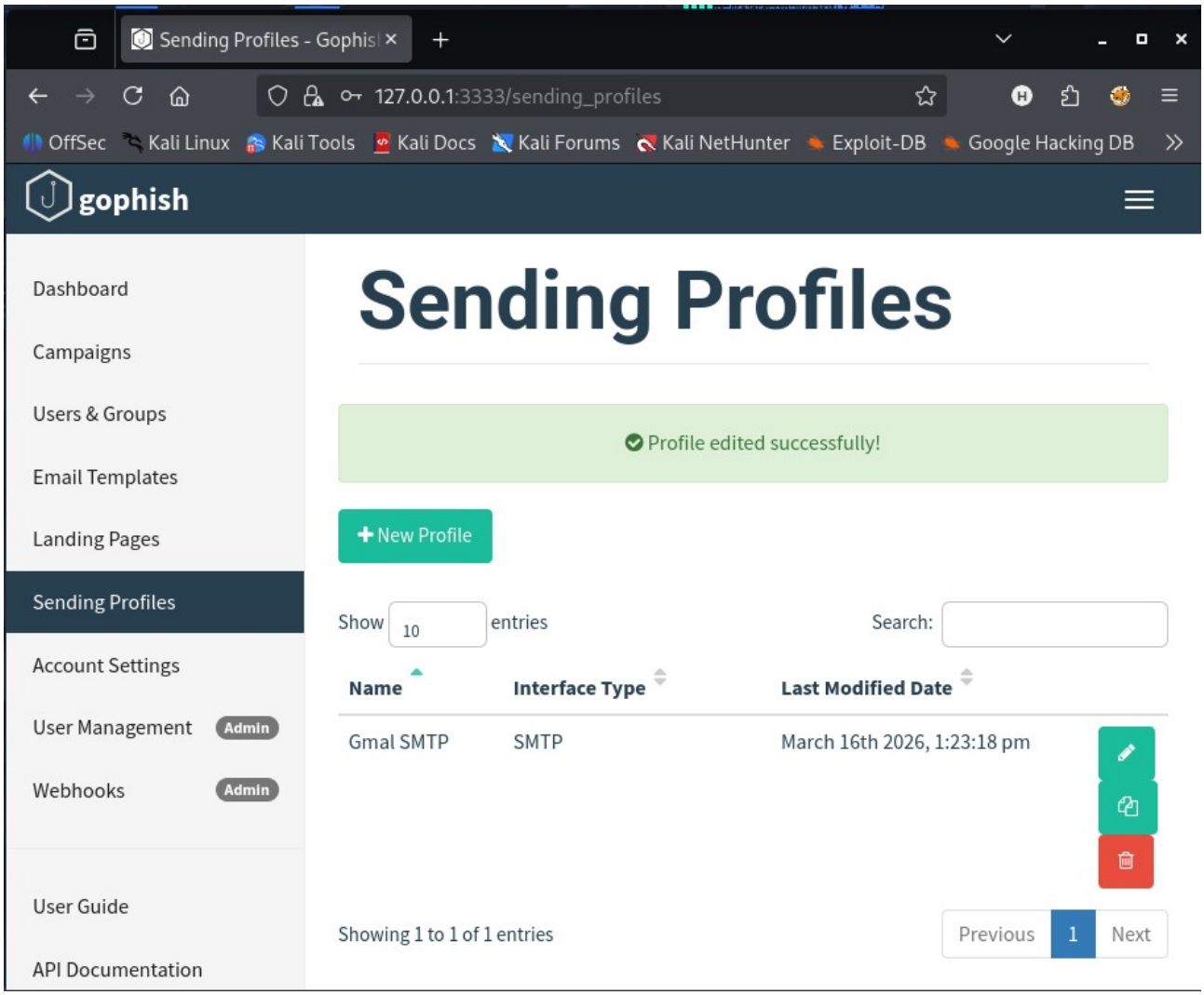
SMTP From
The email address used as the sender.

Host
SMTP server address.

Username
Email account used to authenticate with the SMTP server.

Password
Application password used for authentication.

Once configured, a test email was sent successfully to verify the SMTP configuration.



Creating the Phishing Email Template

An email template was created inside the GoPhish dashboard to simulate a realistic phishing email.

The template was designed to resemble a legitimate security notification sent by an organization.

Example subject line:

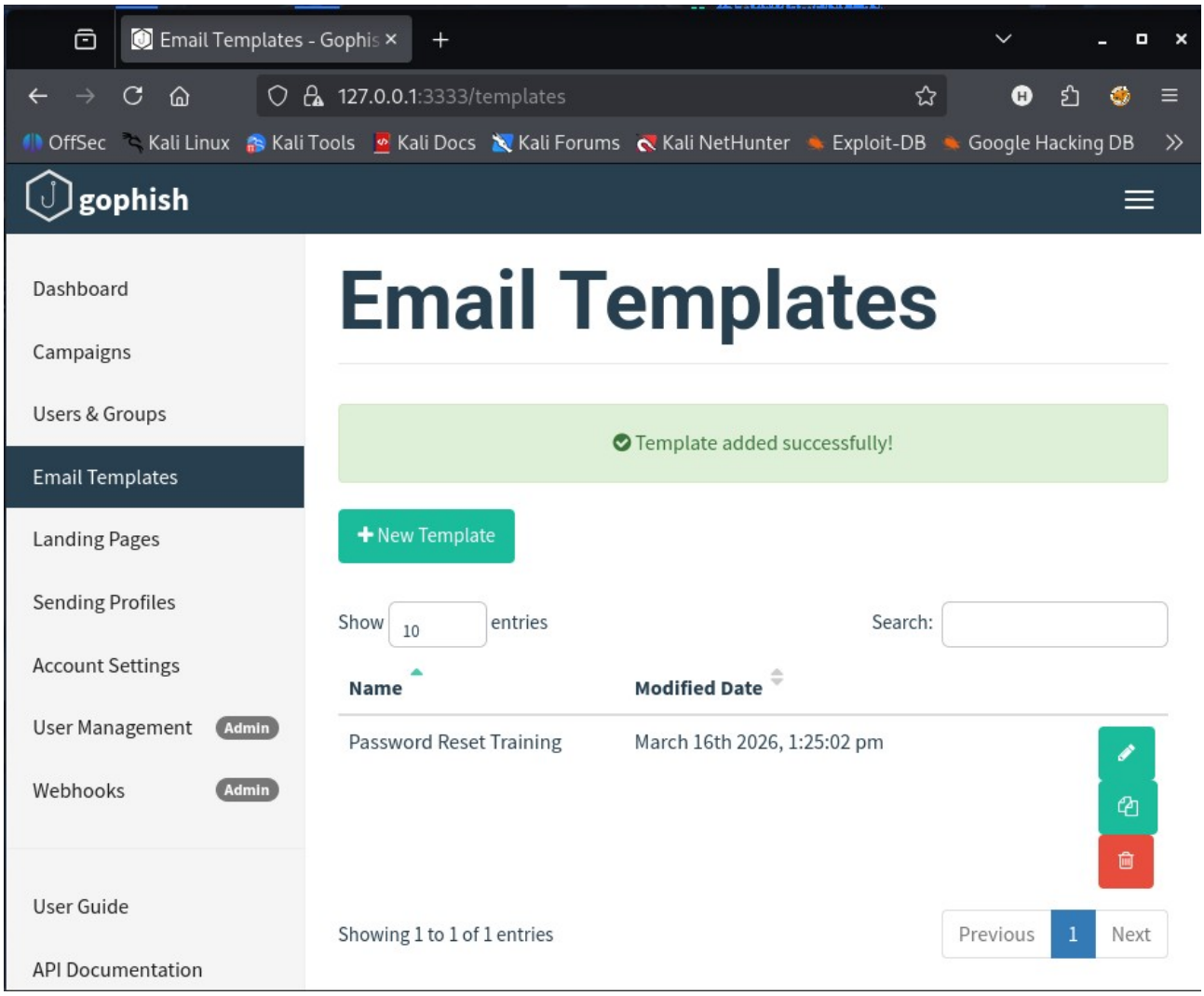
Security Alert - Please Verify Your Account

The email contained a call-to-action button that redirected the user to a login page.

The phishing link generated by GoPhish included a unique tracking identifier to monitor user interaction.

Example:

http://192.168.100.30/?rid=XXXXX



Creating the Phishing Landing Page

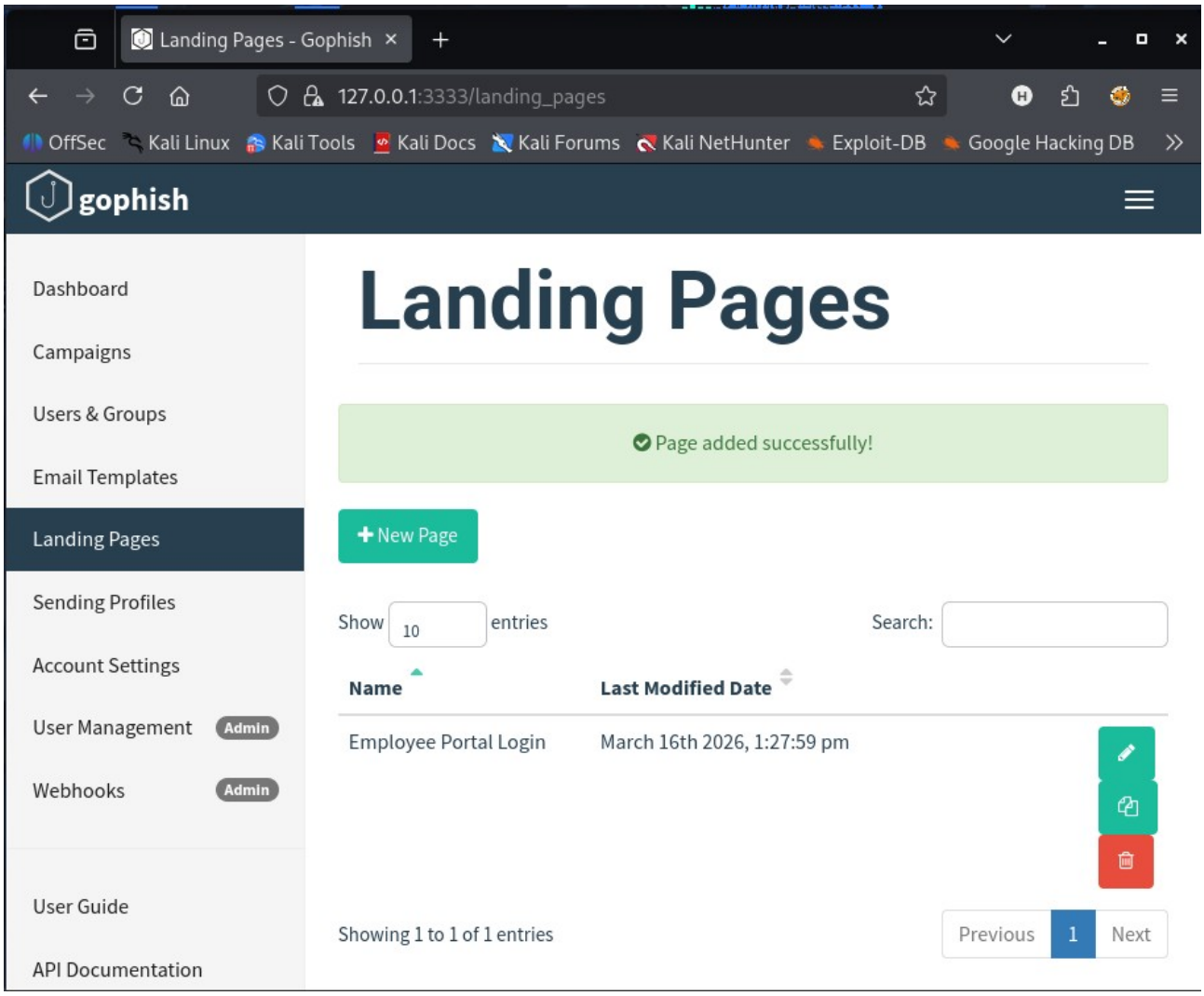
A phishing landing page was created to simulate a login portal where users would enter their credentials.

The page was configured with the following options:

Capture Submitted Data
This option enabled GoPhish to record submitted credentials.

Redirect URL
After submitting credentials, users were redirected to a legitimate website.

The landing page was linked to the phishing email template during the campaign creation process.

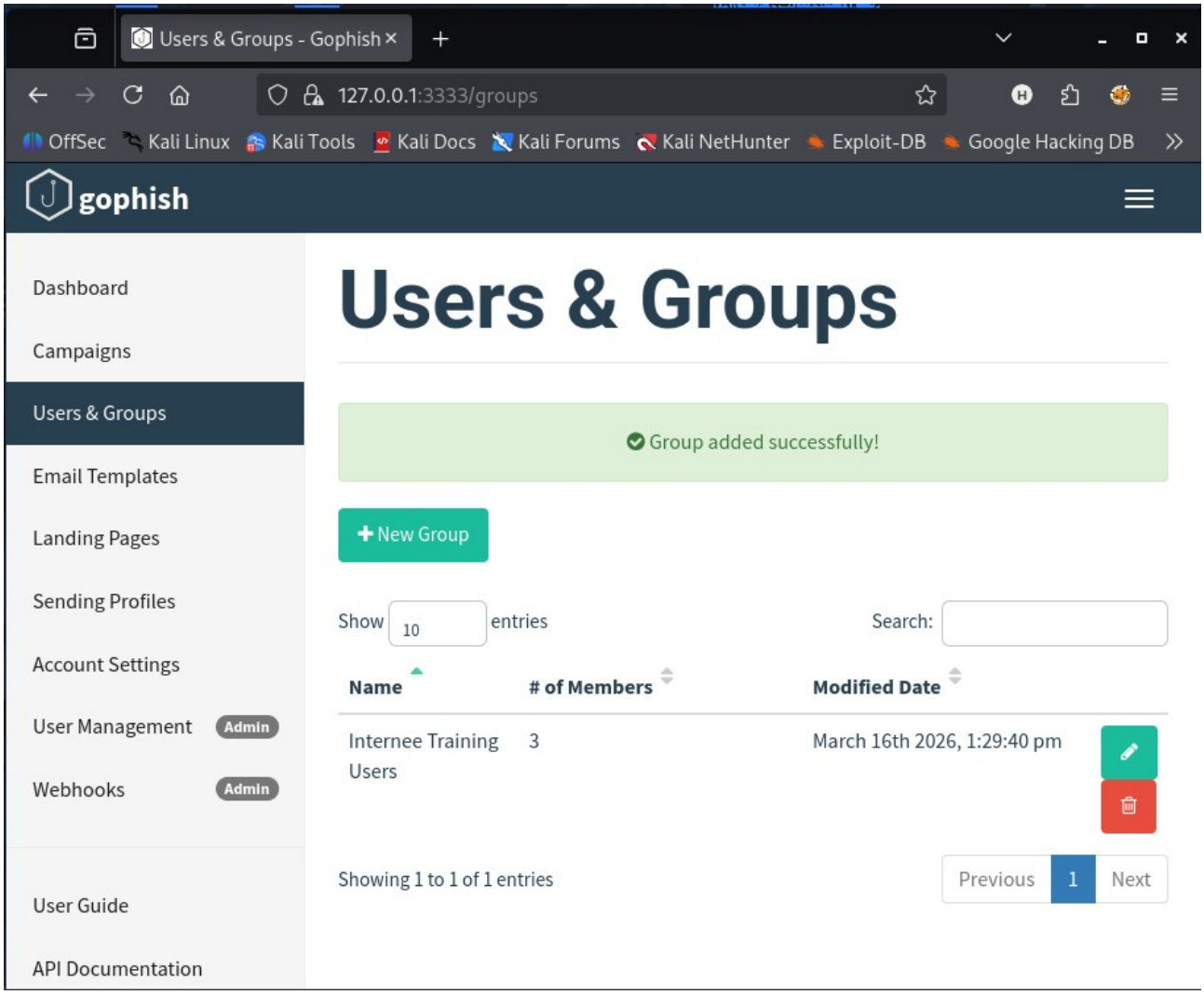


Creating the Target Group

A target group was created in GoPhish to define the recipients of the phishing email.

For this simulation, a test email account was used as the recipient.

This allowed the phishing campaign to be tested in a safe environment without targeting real users.



Launching the Phishing Campaign

A phishing campaign was created using the following components:

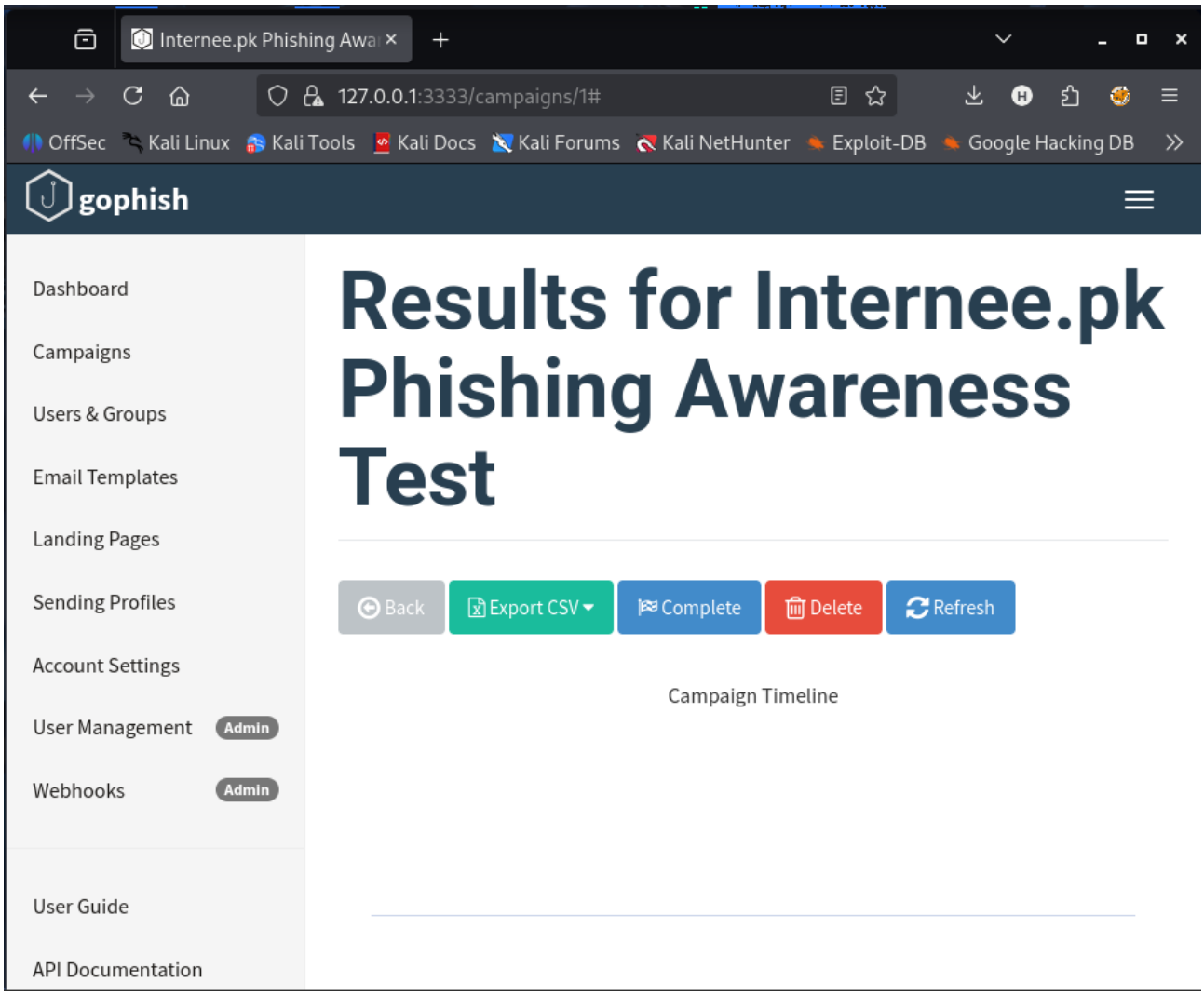
Email Template
Landing Page
Target Group
SMTP Profile

The campaign URL was configured as:

http://192.168.100.30

Once launched, GoPhish generated unique phishing links for each recipient.

These links were embedded in the phishing email and used to track user interaction.



Testing the Campaign

The phishing email was opened on the Windows virtual machine.

The user performed the following actions:

Opened the email
Clicked the phishing link
Entered login credentials on the landing page

These interactions were captured and recorded by the GoPhish dashboard.

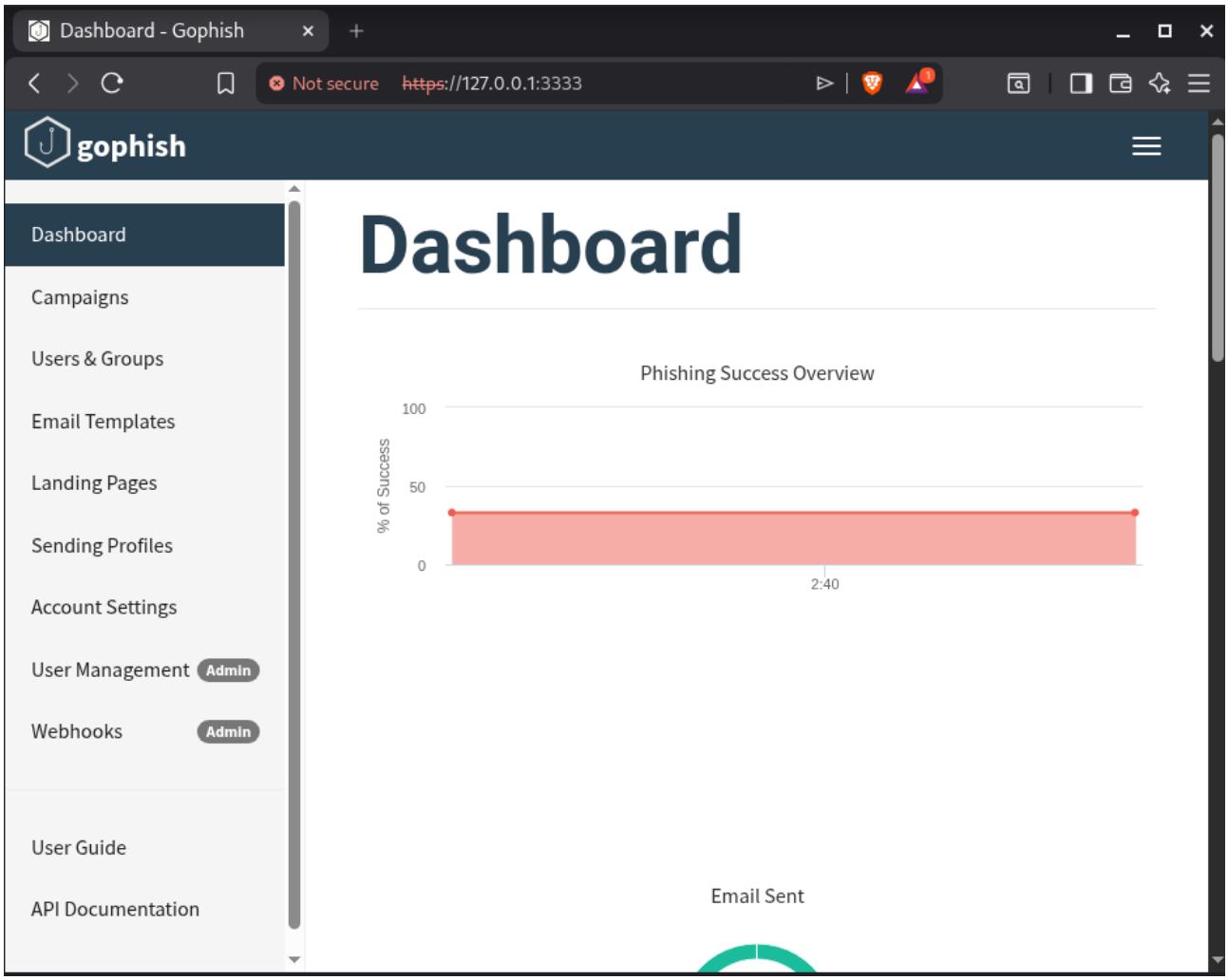
Monitoring Campaign Results

The GoPhish dashboard provides real-time analytics for phishing campaigns.

The following metrics were observed:

Email delivered
Email opened
Link clicked
Credentials submitted

These results demonstrate how organizations can evaluate employee awareness and identify potential security weaknesses.

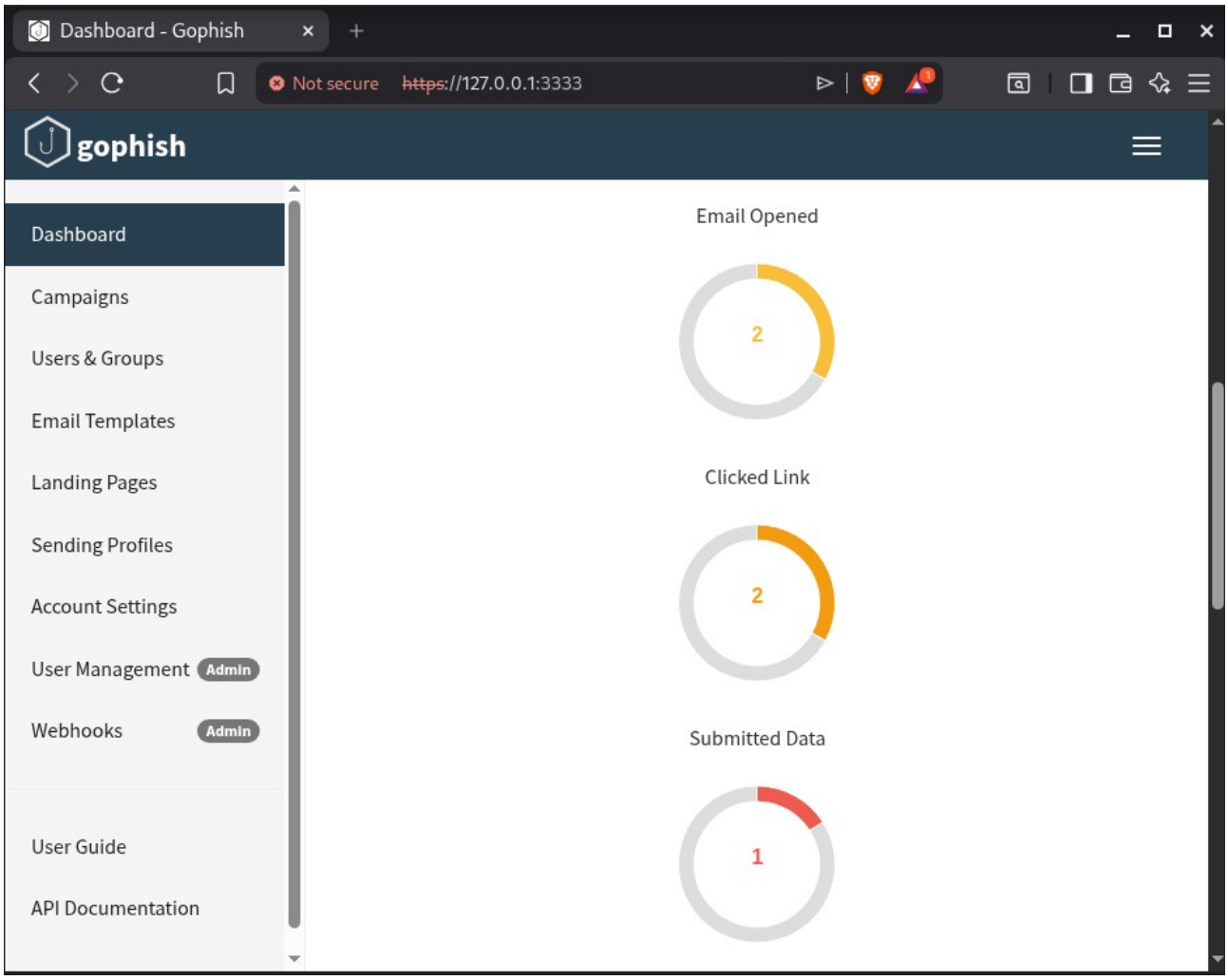


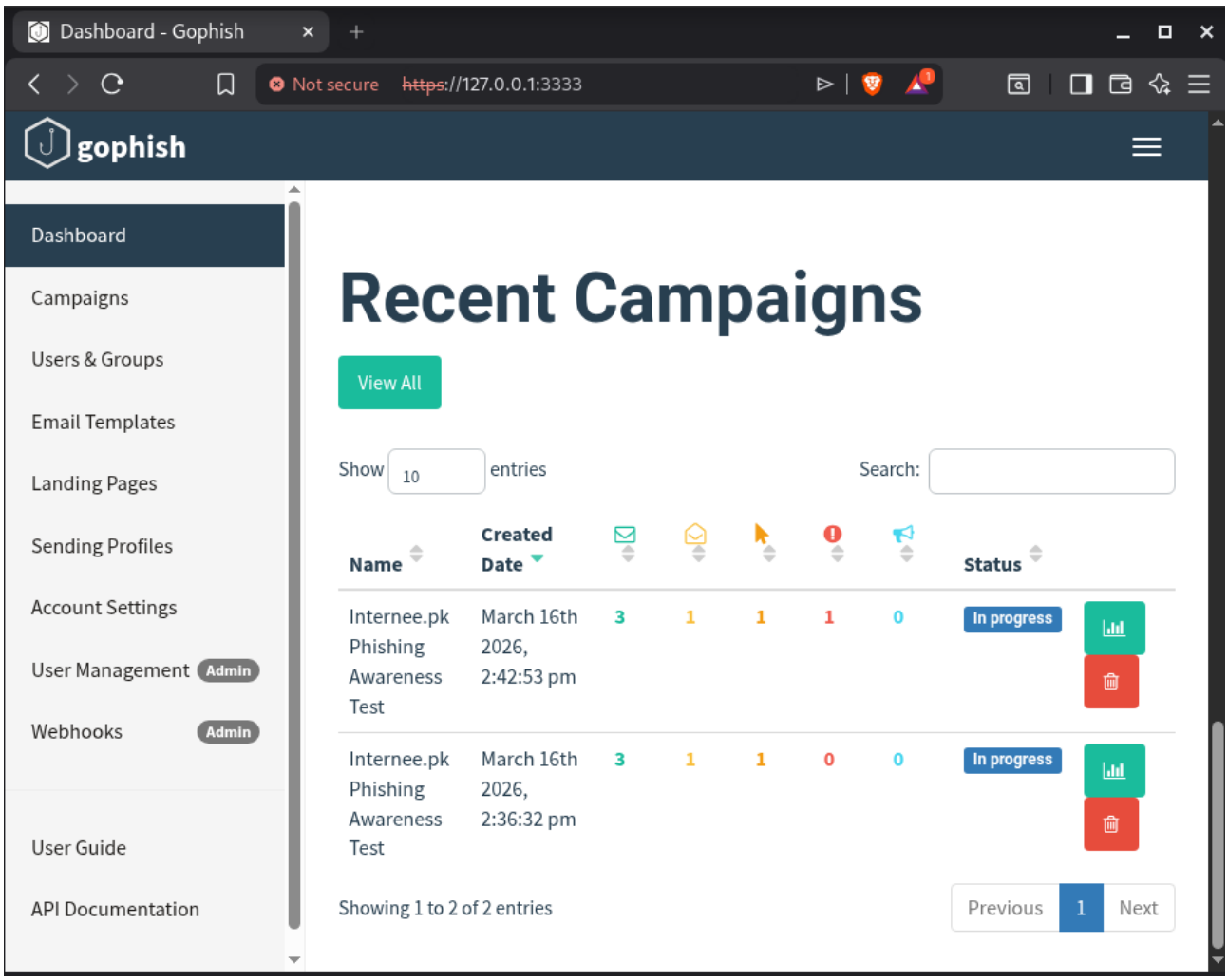
Results and Observations

The phishing simulation successfully demonstrated how attackers can exploit human behavior through deceptive emails.

Even simple phishing emails can trick users into clicking malicious links or revealing sensitive information.

Security teams can use phishing simulations to measure awareness levels and provide targeted training to employees.





Conclusion

Phishing attacks continue to be one of the most effective cyberattack methods used by attackers.

This assignment demonstrated how phishing campaigns are created and monitored using the GoPhish framework.

Through this lab exercise, practical experience was gained in deploying phishing simulations, designing email templates, capturing user interactions, and analyzing campaign results.

Organizations can use phishing awareness training programs to reduce the risk of social engineering attacks and strengthen their overall security posture.

Ethical Disclaimer

This phishing simulation was conducted strictly in a controlled lab environment for educational and training purposes.

No real users or organizations were targeted during this exercise.